1. Install and configure Tomcat version 9 on your Linux machine with the Manager and Host Manager sections enabled.

- **Check Java (Tomcat needs Java)**

```
java -version
```

- **If Java is not installed**

```
sudo apt update
sudo yum install java -y
```

Tomcat runs on Java. Without Java, Tomcat won't start.

- **Create tomcat user**

```
sudo useradd -r -m -U -d /opt/tomcat -s /bin/false tomcat
```

A dedicated non-login user to run Tomcat securely.

- **Install wget**

```
sudo yum install wget -y
```

Needed to download Tomcat from the internet.

- **Download Tomcat**

```
wget https://archive.apache.org/dist/tomcat/tomcat-9/v9.0.98/bin/apache-tomcat-9.0.98.tar.gz
```

Gets the specific Tomcat version archive.

- **Create installation directory**

```
sudo mkdir /opt/tomcat
```

Where Tomcat will be installed.

- **Extract Tomcat**

```
sudo tar xzvf apache-tomcat-9.0.98.tar.gz -C /opt/tomcat
```

Unpacks the downloaded archive into the install directory.

- **Change ownership**

```
sudo chown -R tomcat: /opt/tomcat
```

Gives the tomcat user control over Tomcat files.

- **Make scripts executable**

```
sudo chmod +x /opt/tomcat/bin/*.sh
```

Ensures startup/shutdown scripts can run.

- **Create Tomcat service file**

```
sudo vi /etc/systemd/system/tomcat.service
```

Defines how Tomcat will run as a service.

- **Paste service config**

```
[Unit]
Description=Apache Tomcat Web Application Container
After=network.target
[Service]
Type=forking
User=tomcat
Group=tomcat
Environment="JAVA_HOME=/usr/lib/jvm/java-21-openjdk"
Environment="CATALINA_PID=/opt/tomcat/temp/tomcat.pid"
Environment="CATALINA_HOME=/opt/tomcat" Environment="CATALINA_BASE=/opt/tomcat"
Environment="CATALINA_OPTS=-Xms512M -Xmx1024M -server -XX:+UseParallelGC"
Environment="JAVA_OPTS=-Djava.awt.headless=true
Djava.security.egd=file:/dev/./urandom"
ExecStart=/opt/tomcat/bin/startup.sh
ExecStop=/opt/tomcat/bin/shutdown.sh
Restart=on-failure
[Install]
WantedBy=multi-user.target
```

Configuration for Tomcat user, paths, and options.

- **Reload systemd**

```
sudo systemctl daemon-reload
```

Apply the new service file.

- **Start Tomcat**

```
sudo systemctl start tomcat
```

Launches Tomcat immediately.

- **Enable at boot**

```
sudo systemctl enable tomcat
```

Ensures Tomcat starts on system reboot.

- **Check status**

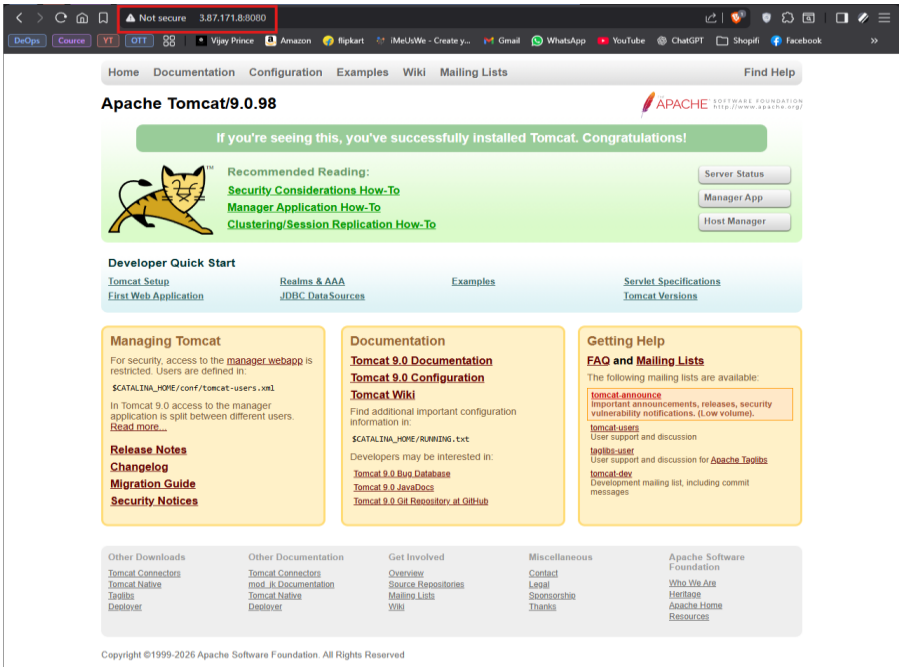
```
sudo systemctl status tomcat
```

Confirms if Tomcat is running.

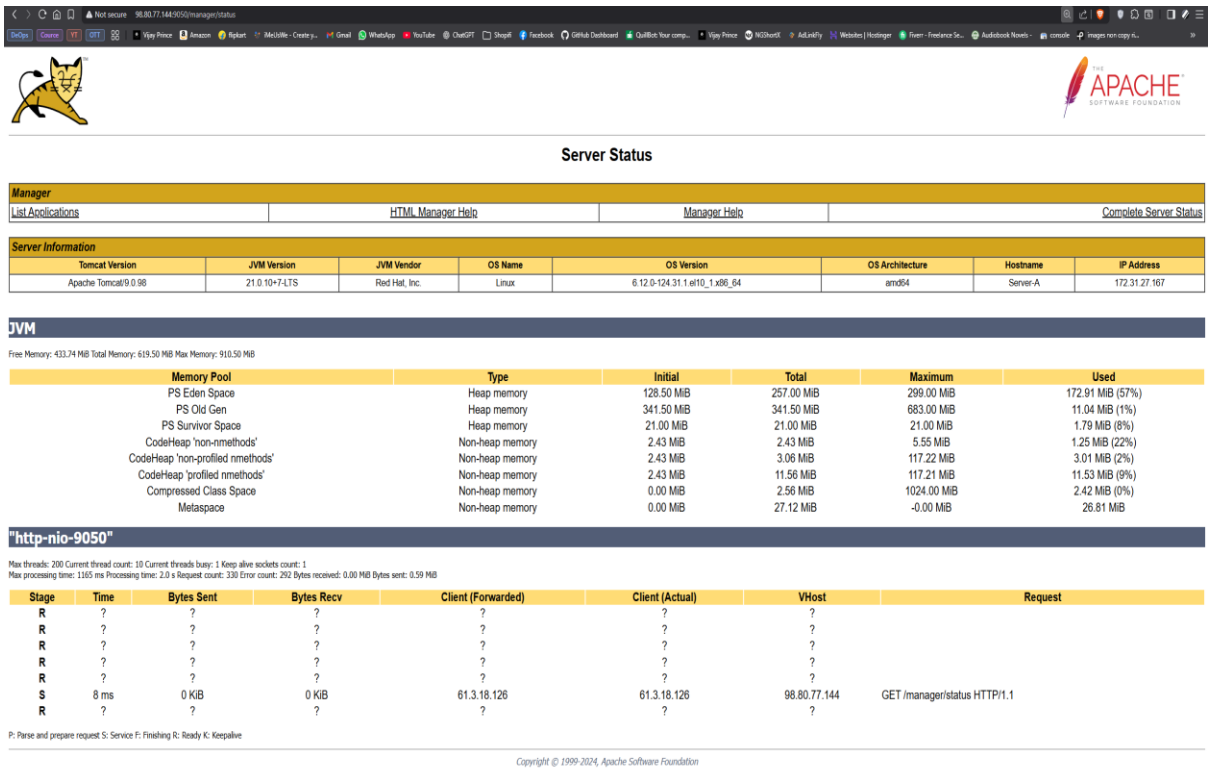
- **Access in browser**

http://3.87.171.8:8080/

Open Tomcat welcome page (default port 8080).



- **Server status**



Server Status

Manager			
List Applications	HTML Manager Help	Manager Help	Complete Server Status

Server Information							
Tomcat Version	JVM Version	JVM Vendor	OS Name	OS Version	OS Architecture	Hostname	IP Address
Apache Tomcat/9.0.98	21.0.10+7-LTS	Red Hat, Inc.	Linux	6.12.0-124.31.1.el10_1.x86_64	amd64	Server-A	172.31.27.167

JVM

Free Memory: 433.74 MiB Total Memory: 619.50 MiB Max Memory: 910.50 MiB

Memory Pool	Type	Initial	Total	Maximum	Used
PS Eden Space	Heap memory	128.50 MiB	257.00 MiB	299.00 MiB	172.91 MiB (57%)
PS Old Gen	Heap memory	341.50 MiB	341.50 MiB	683.00 MiB	11.04 MiB (1%)
PS Survivor Space	Heap memory	21.00 MiB	21.00 MiB	21.00 MiB	1.79 MiB (8%)
CodeHeap 'non-nmethods'	Non-heap memory	2.43 MiB	2.43 MiB	5.55 MiB	1.25 MiB (22%)
CodeHeap 'non-profiled nmethods'	Non-heap memory	2.43 MiB	3.06 MiB	117.22 MiB	3.01 MiB (2%)
CodeHeap 'profiled nmethods'	Non-heap memory	2.43 MiB	11.56 MiB	117.21 MiB	11.53 MiB (9%)
Compressed Class Space	Non-heap memory	0.00 MiB	2.56 MiB	1024.00 MiB	2.42 MiB (0%)
Metaspace	Non-heap memory	0.00 MiB	27.12 MiB	-0.00 MiB	26.81 MiB

"http-nio-9050"

Max threads: 200 Current thread count: 10 Current threads busy: 1 Keep alive sockets count: 1
Max processing time: 1165 ms Processing time: 2.0 s Request count: 330 Error count: 292 Bytes received: 0.00 MiB Bytes sent: 0.59 MiB

Stage	Time	Bytes Sent	Bytes Recv	Client (Forwarded)	Client (Actual)	VHost	Request
R	?	?	?	?	?	?	
R	?	?	?	?	?	?	
R	?	?	?	?	?	?	
R	?	?	?	?	?	?	
R	?	?	?	?	?	?	
S	8 ms	0 KiB	0 KiB	61.3.18.126	61.3.18.126	98.80.77.144	GET /manager/status HTTP/1.1
R	?	?	?	?	?	?	

P: Parse and prepare request S: Service F: Finishing R: Ready K: Keepalive

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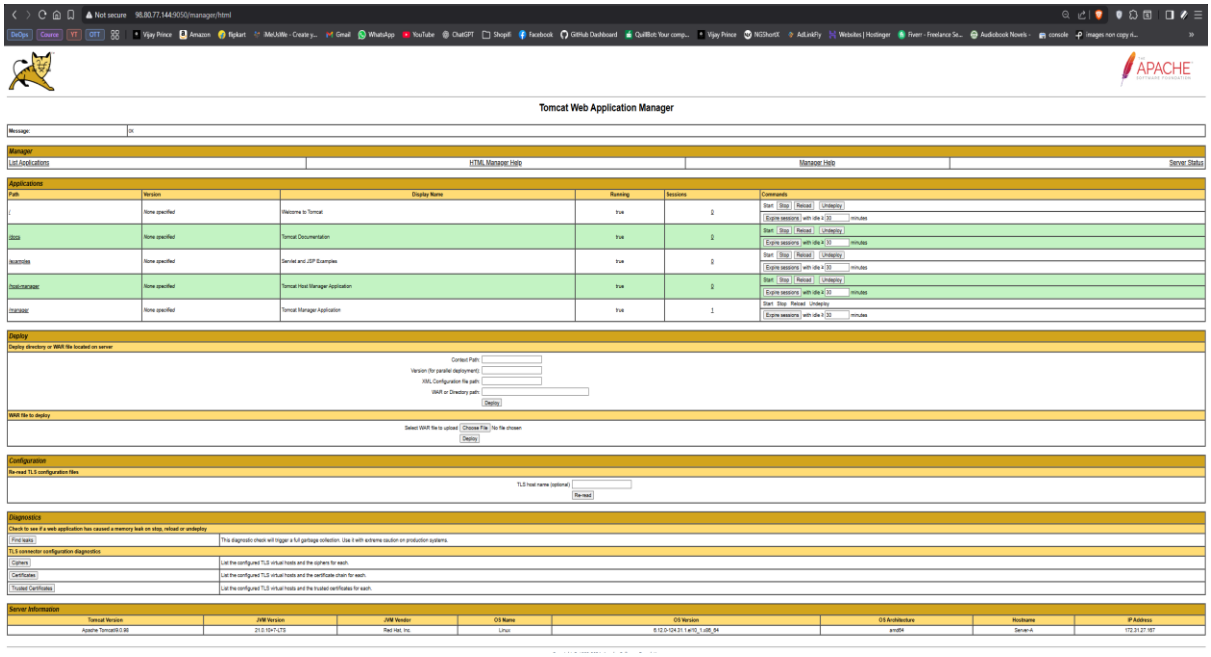
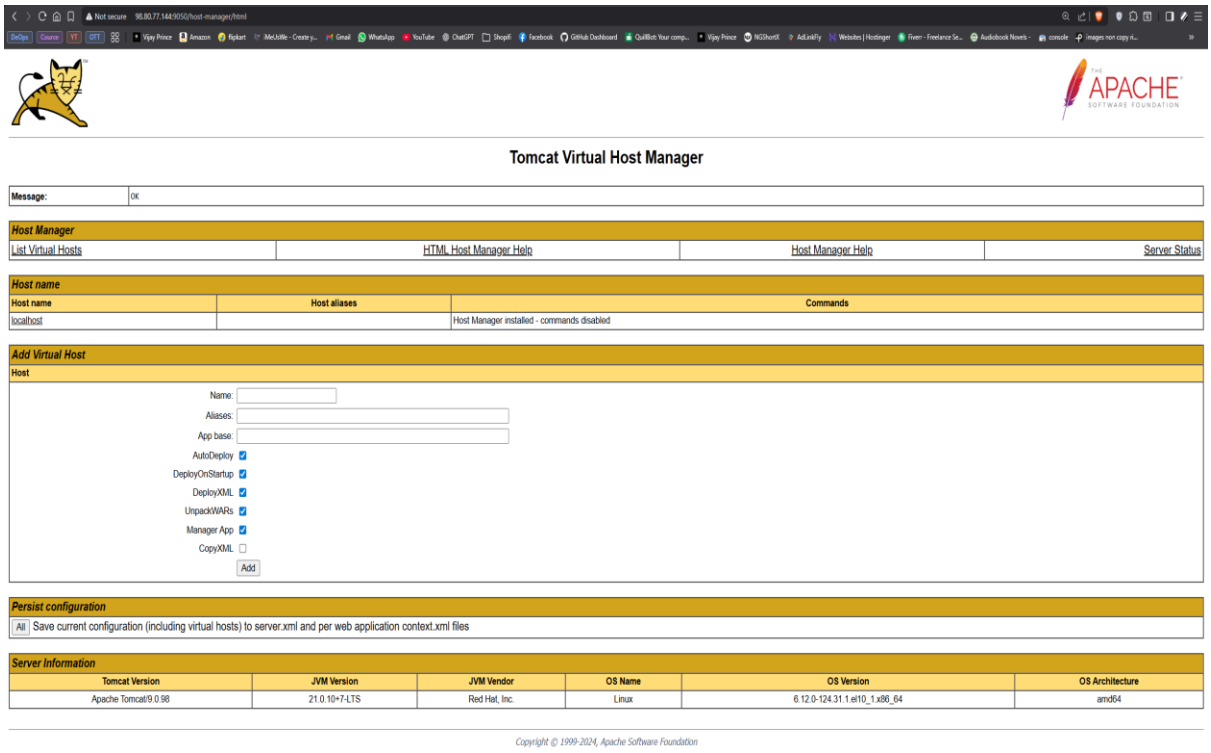
- **Manager App**



- **Host Manager**



2. Change the default Tomcat port to 9050.

- **Edit Tomcat Port**

```
sudo vi /opt/tomcat/conf/server.xml
```

Open the main Tomcat config file.

- **Change port**

```
:%s/8080/9050/g
```

Replace all 8080 with 9050.

- **Save & quit**

```
:wq
```

Write changes and exit editor.

- **Restart Tomcat**

```
sudo systemctl restart tomcat
```

Apply the new port configuration.

- **Check listening port**

```
sudo ss -tulnp | grep 9050
```

Verify Tomcat is running on 9050.

- **Open firewall port**

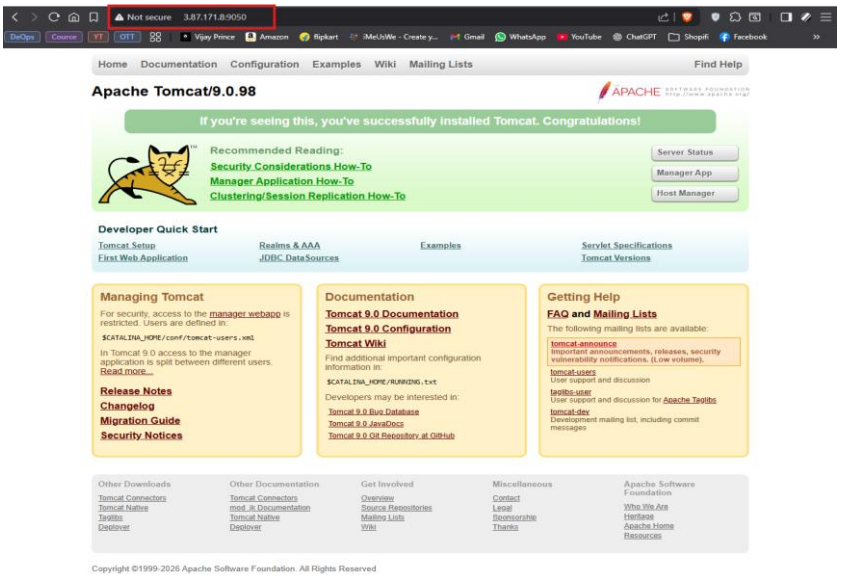
```
sudo firewall-cmd --permanent --add-port=9050/tcp
sudo firewall-cmd --reload
```

Allow external access to 9050.

- **Access in browser**

```
http://3.87.171.8:9050/
```

Use new port 9050.



3. Create a new user and configure the necessary settings for secure login via Putty or other remote access tools.

- **Connect to EC2**

```
# AWS Machine (Linux)→ ssh -i zoro.pem ec2-user@3.87.171.8
```

Login as default user using your key.

- **Create new user**

```
sudo useradd devivaraprasad
```

Adds a new system user for secure login.

- **Set password (optional)**

```
sudo passwd devivaraprasad
```

Assigns password if needed (not required for key login).

- **Give sudo permissions**

```
# Amazon Linux / CentOS → sudo usermod -aG wheel devivaraprasad
```

Allows the new user to run admin commands.

- **Create .ssh directory**

```
sudo mkdir /home/devivaraprasad/.ssh
sudo chmod 700 /home/devivaraprasad/.ssh
```

Stores authorized keys for the new user.

- **Copy authorized keys**

```
sudo cp /home/ec2-user/.ssh/authorized_keys /home/devivaraprasad/.ssh/
```

Reuses existing key from ec2-user.

- **Set correct permissions**

```
sudo chown -R devivaraprasad:devivaraprasad /home/devivaraprasad/.ssh
sudo chmod 600 /home/devivaraprasad/.ssh/authorized_keys
```

Ensures only devivaraprasad can access SSH keys.

- **Test SSH login**

```
# Linux / Mac→ ssh -i zoro.pem devivaraprasad@3.87.171.8
```

Connect as new user using same key.

- **Windows Subsystem for Linux→** Host: devivaraprasad@3.87.171.8, select existing .pem key.

```
ec2-user@ip-172-31-27-167:~
[ec2-user@ip-172-31-27-167 ~]$ sudo useradd devivaraprasad^C
[ec2-user@ip-172-31-27-167 ~]$ sudo usermod -aG wheel devivaraprasad
[ec2-user@ip-172-31-27-167 ~]$ sudo mkdir /home/devivaraprasad/.ssh
[ec2-user@ip-172-31-27-167 ~]$ sudo chmod 700 /home/devivaraprasad/.ssh
[ec2-user@ip-172-31-27-167 ~]$ sudo cp /home/ec2-user/.ssh/authorized_keys /home/ devivaraprasad/.ssh/
```

```
[ec2-user@ip-172-31-27-167 ~]$ sudo cp /home/ec2-user/.ssh/authorized_keys /home/devivaraprasad/.ssh/
[ec2-user@ip-172-31-27-167 ~]$ sudo chown -R devivaraprasad:devivaraprasad /home/devivaraprasad/.ssh
[ec2-user@ip-172-31-27-167 ~]$ sudo chmod 600 /home/devivaraprasad/.ssh/authorized_keys
[ec2-user@ip-172-31-27-167 ~]$ ssh -i zoro.pem devivaraprasad@3.87.171.8
Register this system with Red Hat Insights: rhc connect

Example:
# rhc connect --activation-key <key> --organization <org>

The rhc client and Red Hat Insights will enable analytics and additional
management capabilities on your system.
View your connected systems at https://console.redhat.com/insights

You can learn more about how to register your system
using rhc at https://red.ht/registration
[devivaraprasad@ip-172-31-27-167 ~]$
```

4. Establish passwordless connection between two linux servers

Server A (remote /source): 3.87.171.8 (your EC2 server) where you run ssh-keygen

Server B (remote/target): 34.229.202.93 (your EC2 server)

- **Generate SSH key**

```
ssh-keygen -t rsa
```

Creates a new RSA key pair for passwordless login.

- **Verify keys**

```
ls -l ~/.ssh/
```

Check that private and public keys are created.

You should see files like:

id_rsa (private key)

id_rsa.pub (public key)

- **View public key:**

```
cat ~/.ssh/id_rsa.pub
```

- **Append to remote target server**

echo “” >> .ssh/authorized_keys (in **34.229.202.93**)

Add public key to authorized_keys.

- **Test passwordless login**

```
ssh ubuntu@34.229.202.93  (in 3.87.171.8)
```

Connect from Server A without password

```
[ec2-user@Server-A ~]$ ssh ubuntu@34.229.202.93
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-1018-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Mon Feb  9 15:27:02 UTC 2026

  System load:  0.0               Temperature:           -273.1 C
  Usage of /:   28.9% of 6.71GB   Processes:             116
  Memory usage: 25%               Users logged in:       1
  Swap usage:   0%                IPv4 address for ens5: 172.31.22.145


Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


The list of available updates is more than a week old.
To check for new updates run: sudo apt update

Last login: Mon Feb  9 15:16:03 2026 from 117.205.168.152
ubuntu@Server-B:~$
```

5. Create a new user account on your Linux machine and grant them sudo privileges.

- **Create new user**

```
sudo useradd newuser
```

Adds a new system user.

- **Add user to sudo group**

```
sudo usermod -aG sudo newuser # Ubuntu / Debian
sudo usermod -aG wheel newuser # CentOS / RHEL / Amazon Linux
```

Gives admin privileges.

- **Verify sudo access**

```
su  newuser
```

Switch to new user.

- **Test sudo**

```
sudo whoami
```

Run a command as sudo to confirm privileges.

- **Test passwordless login**

```
ssh ubuntu@34.229.202.93  (in 3.87.171.8)
```

Connect from Server A without password

Output should be: **root**

```
[ec2-user@Server-A home]$ sudo passwd prasad
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: password updated successfully
[ec2-user@Server-A home]$ sudo su prasad
[prasad@Server-A home]$ sudo whoami

We trust you have received the usual lecture from the local System
Administrator. It usually boils down to these three things:

    #1) Respect the privacy of others.
    #2) Think before you type.
    #3) With great power comes great responsibility.

For security reasons, the password you type will not be visible.

[sudo] password for prasad:
root
[prasad@Server-A home]$
```

6. Use the 'scp' command to securely copy files and directories between Linux machines without password

- **Copy a single file**

```
scp f9.sh ubuntu@34.229.202.93:/home/ubuntu/
```

Securely transfer a file to a remote server.

➔ Copies **f9.sh** to /home/ubuntu/ → on remote server.

- **Copy entire directory**

```
scp -r king ubuntu@34.229.202.93:/home/ubuntu/
```

Transfer a folder recursively.

➔ Copies the full **king**/ directory and all contents to remote server.

```
[ec2-user@Server-A ~]$ ls
a1                              aws     f2      id_rsa.pub      narendra
a2                              aws.sh  f3      id_rsa.pub.pub  table
apache-tomcat-9.0.98.tar.gz     c1      f4.txt  king            ubuntu@34.229.202.93
apache-tomcat-9.0.98.tar.gz.1   d1      f9.sh   l1              zoro.pem
[ec2-user@Server-A ~]$ scp f9.sh ubuntu@34.229.202.93:/home/ubuntu/
f9.sh                                                     100% 1271     2.4MB/s   00:00
[ec2-user@Server-A ~]$ scp -r king ubuntu@34.229.202.93:/home/ubuntu/
t2                                                        100%    0     0.0KB/s   00:00
t3                                                        100%    0     0.0KB/s   00:00
t4                                                        100%    0     0.0KB/s   00:00
t5                                                        100%    0     0.0KB/s   00:00
original.sh                                               100%    0     0.0KB/s   00:00
t1                                                        100%   59    18.6KB/s   00:00
fi                                                        100%    0     0.0KB/s   00:00
f1                                                        100%   79    74.3KB/s   00:00
f2                                                        100%    0     0.0KB/s   00:00
f3                                                        100%    0     0.0KB/s   00:00
f4                                                        100%    0     0.0KB/s   00:00
bestfriend                                                100%    0     0.0KB/s   00:00
f1.txt                                                    100%    0     0.0KB/s   00:00
[ec2-user@Server-A ~]$
```

```
ubuntu@Server-B:~$ ls
f9.sh  king  password.pem
ubuntu@Server-B:~$
```

7. How can you list the contents of a tar.gz file and extract a particular file

- **List contents**

```
tar -tzf apache-tomcat-9.0.98.tar.gz
```

View all files inside a tar.gz archive without extracting.

- Shows detailed list of all files in the archive.

- **List contents**

```
tar -xzf apache-tomcat-9.0.98.tar.gz apache-tomcat-9.0.98/conf/server.xml
```

Extract only a specific file from the archive.

- Extracts path/to/file from the tar.gz archive without unpacking everything.

8. How can you compress a directory or file into a tar archive on Linux

- **Compress a directory**

```
tar -czf backup.tar.gz /home/ec2-user/king/
```

Create a tar.gz archive from a folder.

➔ Compresses tomcat/ directory into king.tar.gz.

```
[ec2-user@Server-A ~]$ ls
a1                              aws     f3              king                    zoro.pem
a2                              aws.sh  f4.txt          l1
apache-tomcat-9.0.98            c1      f9.sh           narendra
apache-tomcat-9.0.98.tar.gz     d1      id_rsa.pub      table
apache-tomcat-9.0.98.tar.gz.1   f2      id_rsa.pub.pub  ubuntu@34.229.202.93
[ec2-user@Server-A ~]$ tar -czf backup.tar.gz king/
[ec2-user@Server-A ~]$ ls
a1                              aws             f2          id_rsa.pub.pub  ubuntu@34.229.202.93
a2                              aws.sh          f3          king            zoro.pem
apache-tomcat-9.0.98            backup.tar.gz   f4.txt      l1
apache-tomcat-9.0.98.tar.gz     c1              f9.sh       narendra
apache-tomcat-9.0.98.tar.gz.1   d1              id_rsa.pub  table
[ec2-user@Server-A ~]$
```

- **Compress multiple files**

```
tar -czf archive.tar.gz file1 file2 file3
```

Include specific files in one archive.

➔ Compresses file1, file2, file3 into archive.tar.gz.

```
[ec2-user@Server-A ~]$ ls
a1                              aws             f2          id_rsa.pub.pub  ubuntu@34.229.202.93
a2                              aws.sh          f3          king            zoro.pem
apache-tomcat-9.0.98            backup.tar.gz   f4.txt      l1
apache-tomcat-9.0.98.tar.gz     c1              f9.sh       narendra
apache-tomcat-9.0.98.tar.gz.1   d1              id_rsa.pub  table
[ec2-user@Server-A ~]$ tar -czf backup_files.tar.gz f2 f3 f4.txt
[ec2-user@Server-A ~]$ ls
a1                              aws.sh              f3              l1
a2                              backup.tar.gz       f4.txt          narendra
apache-tomcat-9.0.98            backup_files.tar.gz f9.sh           table
apache-tomcat-9.0.98.tar.gz     c1                  id_rsa.pub      ubuntu@34.229.202.93
apache-tomcat-9.0.98.tar.gz.1   d1                  id_rsa.pub.pub  zoro.pem
aws                             f2                  king
[ec2-user@Server-A ~]$
```

- **List contents:** tar -tzf archive.tar.gz
- **Extract one file:** tar -xzf archive.tar.gz path/to/file
- **Compress:** tar -czf archive.tar.gz

9. Write a command to synchronise files and directories between 2 local folders

- **Basic sync** → One-way sync without deleting extra files.

```
rsync -av /home/user/folder1/ /home/user/folder2/
```

Synchronises changes from folder1 to folder2 without removing extra files.

```
[ec2-user@Server-A ~]$ ls
a1                              aws                 d1      id_rsa.pub      table
a2                              aws.sh              f2      id_rsa.pub.pub  ubuntu@34.229.202.93
apache-tomcat-9.0.98            backup.tar.gz       f3      king            zoro.pem
apache-tomcat-9.0.98.tar.gz     backup_files.tar.gz f4.txt  l1
apache-tomcat-9.0.98.tar.gz.1   c1                  f9.sh   narendra
[ec2-user@Server-A ~]$ rsync -av /home/ec2-user/king /home/ec2-user/king1
sending incremental file list
created directory /home/ec2-user/king1
king/
king/f1.txt
king/naga/
king/naga/bestfriend
king/naga/f1
king/naga/f2
king/naga/f3
king/naga/f4
king/naga/fi
king/naga/tata/
king/naga/tata/t1
king/naga/tata/t2
king/naga/tata/t3
king/naga/tata/t4
king/naga/tata/t5
king/naga/tata/tata1/
king/naga/tata/tata1/original.sh

sent 1,102 bytes  received 330 bytes  2,864.00 bytes/sec
total size is 138  speedup is 0.10
[ec2-user@Server-A ~]$ ls
a1                              aws                 d1      id_rsa.pub      narendra
a2                              aws.sh              f2      id_rsa.pub.pub  table
apache-tomcat-9.0.98            backup.tar.gz       f3      king            ubuntu@34.229.202.93
apache-tomcat-9.0.98.tar.gz     backup_files.tar.gz f4.txt  king1           zoro.pem
apache-tomcat-9.0.98.tar.gz.1   c1                  f9.sh   l1
[ec2-user@Server-A ~]$
```

- **Mirror sync** → Exact copy, deletes files not in source.

```
rsync -av --delete /home/user/folder1/ /home/user/folder2/
```

Ensures folder2 exactly matches folder1.

```
[ec2-user@Server-A ~]$ ls
a1                              aws                 d1      id_rsa.pub      narendra
a2                              aws.sh              f2      id_rsa.pub.pub  ramu
apache-tomcat-9.0.98            backup.tar.gz       f3      king            table
apache-tomcat-9.0.98.tar.gz     backup_files.tar.gz f4.txt  king1           ubuntu@34.229.202.93
apache-tomcat-9.0.98.tar.gz.1   c1                  f9.sh   l1              zoro.pem
[ec2-user@Server-A ~]$ rsync -av --delete /home/ec2-user/king /home/ec2-user/ramu
sending incremental file list
king/
king/f1.txt
king/naga/
king/naga/bestfriend
king/naga/f1
king/naga/f2
king/naga/f3
king/naga/f4
king/naga/fi
king/naga/tata/
king/naga/tata/t1
king/naga/tata/t2
king/naga/tata/t3
king/naga/tata/t4
king/naga/tata/t5
king/naga/tata/tata1/
king/naga/tata/tata1/original.sh

sent 1,102 bytes  received 287 bytes  2,778.00 bytes/sec
total size is 138  speedup is 0.10
[ec2-user@Server-A ~]$ ls
a1                              aws                 d1      id_rsa.pub      narendra
a2                              aws.sh              f2      id_rsa.pub.pub  ramu
apache-tomcat-9.0.98            backup.tar.gz       f3      king            table
apache-tomcat-9.0.98.tar.gz     backup_files.tar.gz f4.txt  king1           ubuntu@34.229.202.93
apache-tomcat-9.0.98.tar.gz.1   c1                  f9.sh   l1              zoro.pem
[ec2-user@Server-A ~]$
```

Quick flags summary →

- -a → archive mode (preserves permissions, timestamps, symlinks)
- -v → verbose (shows progress)
- --delete → remove files from destination not present in source

10. What is load average and swap memory in Linux

Load Average → shows the average number of processes that are running or waiting for CPU (and uninterruptible I/O) in the system.

uptime → Used to check load average

```
[ec2-user@Server-A ~]$ uptime
 13:10:10 up 46 min,  4 users,  load average: 0.00, 0.00, 0.00
[ec2-user@Server-A ~]$
```

➞ The three load average values represent system load for 1, 5 and 15 minutes respectively.

Swap Memory → is disk space used by Linux as virtual memory when physical RAM is not sufficient.

It helps prevent applications from being terminated when RAM becomes full.

free -h

```
[ec2-user@Server-A ~]$ free -h
              total        used        free      shared  buff/cache   available
Mem:           907Mi       598Mi       145Mi       5.3Mi       288Mi       308Mi
Swap:             0B          0B          0B
[ec2-user@Server-A ~]$
```

The Swap row shows the total swap space, used swap and free swap on the system.

11. What are the important system directories in linux and what are they genereally used for [Ex: var, home etc]

Root directory → Base of the filesystem.

/

Essential user binaries → Commands like ls, cp.

/bin

System binaries → Admin commands like ifconfig, fsck.

/sbin

Boot files → Boot loader, Linux kernel, initramfs.

/boot

Device files → Hardware devices like /dev/sda.

/dev

System configuration → Stores configuration files.

/etc

User home directories → Each user's personal files.

/home

Libraries → Shared libraries for binaries.

/lib

Variable data → Logs, spool files, mail, etc.

/var

Temporary files → Cleared on reboot.

/tmp

User programs/data → Contains /usr/bin, /usr/lib.

/usr

Optional software → Third-party packages.

/opt

Root home → Home directory for root user.

/root

Temporary mount points → For disks mounted temporarily.

/mnt

Removable media → USB, CD-ROM mount points.

/media

Service data → Data for services like HTTP, FTP.

/srv

12. What are the package installers for Redhat, Ubuntu, Debian, CentOS, Alpine

RedHat / CentOS → Uses YUM or DNF for package management.

sudo yum install package-name # **Older systems**

sudo dnf install package-name # **Newer systems**

Ubuntu / Debian → Uses APT (Advanced Package Tool).

sudo apt update

sudo apt install package-name

Alpine Linux → Uses APK (Alpine Package Keeper).

sudo apk add package-name